Choose axes that help explain the ramp

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Weight components / ES: componentes del peso / PT: componentes do peso

mg sin θ along; mg cos θ into the ramp.

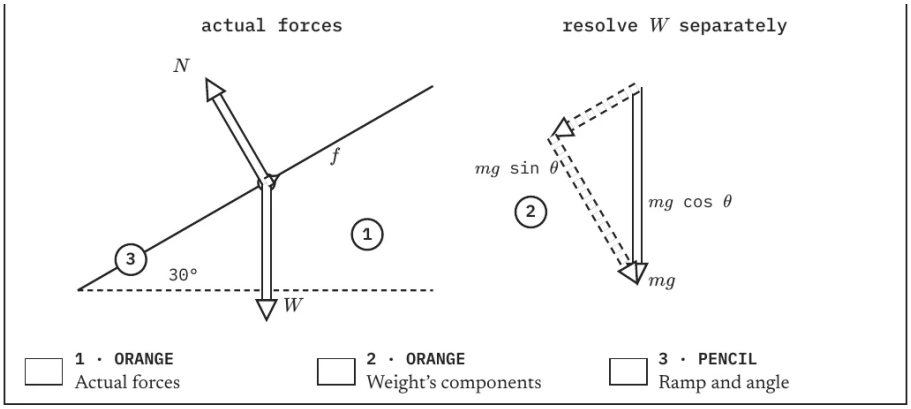
Also on your note page: Parallel axis, Perpendicular axis, Constraint.

B READ THE RELATIONSHIP

Downhill: ma = mg sin θ − f. Perpendicular: N = mg cos θ here.

As you read, find: parallel · perpendicular · downhill.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A 2 kg block slides down a 30° ramp with 2 N friction uphill. Find its acceleration and normal force. Use g = 9.8 m/s² .

Given m = 2 kg ; θ = 30° f = 2 N uphill a ⊥ = 0

Rule mg sin θ − f = ma ∥ N = mg cos θ

Work a ∥ = [19.6(0.5) − 2]/2 N = 19.6 cos 30°

Answer 3.9 m/s² Downhill; N ≈ 17.0 N .

E SAVE FOR THE FINAL MEETING

A suitable set of axes can simplify the forces changing XJ's path. Components are calculation tools; they must not be counted again as extra forces.

Read the physics: Weight components

SUPPORTED READING / read one part, then point to the evidence Name

01 PICTURE IT

On a ramp, gravity still points straight down. The normal force points straight out of the surface. Choose one axis along the ramp and one perpendicular to it.

02 HOW IT WORKS

Split weight into two parts. The parallel part pulls downhill; the perpendicular part presses into the ramp. Calculate with the parts or with the original weight, never with both together.

03 WHAT TO CHECK

Subtract uphill friction from the downhill force component. If the block stays on the ramp, its acceleration perpendicular to the ramp is zero.

04 TRY THE EXAMPLE FROM YOUR NOTES

A 2 kg block slides down a 30° ramp with 2 N friction uphill. Find its acceleration and normal force. Use $g = 9.8\ m/s^2$. 3.9 m/s^2 Downhill; $N \approx 17.0$ N .

Model check: Downhill: $ma = mg \sin\theta - f$. Perpendicular: $N = mg \cos\theta$ here.

05 CONNECT TO XJ

A suitable set of axes can simplify the forces changing XJ's path. Components are calculation tools; they must not be counted again as extra forces.

SAY IT BEFORE YOU WRITE

Why would adding both weight and its components be a mistake? Along the ramp I use ___. Across the ramp I use ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Parallel axis

ES: eje paralelo / PT: eixo paralelo

Meaning in my words	How can you tell?
How would you explain parallel axis to a friend? In my words, this means ___.	If downhill is positive, what sign does the along-ramp part of gravity have? I can tell because ___.
Example or sketch	**Non-example and why**
Sketch axes for a ramp and mark the parallel axis. My example shows ___.	A partner uses the horizontal floor as the parallel axis. Which surface should they follow? I would change ___ because ___.

Perpendicular axis

ES: eje perpendicular / PT: eixo perpendicular

Meaning in my words	How can you tell?
How would you explain perpendicular axis to a friend? In my words, this means ___.	If the positive axis points away from the ramp, what sign does the normal force have? I can tell because ___.
Example or sketch	**Non-example and why**
Draw the perpendicular axis for a block on a tilted ramp. My example shows ___.	A student draws that axis straight up on a tilted ramp. Why would you rotate it? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Weight components

ES: componentes del peso / PT: componentes do peso

Meaning in my words	How can you tell?
What are the two components of weight on a ramp? In my words, this means ___.	Which part of weight pulls downhill, and which presses into the ramp? I can tell because ___.
Example or sketch	**Non-example and why**
Draw weight and its two parts. Which arrows should enter the two ramp-axis equations? My example shows ___.	A student adds weight and both of its components as three forces. What got counted twice? I would change ___ because ___.

Constraint

ES: restricción / PT: restrição

Meaning in my words	How can you tell?
How would you explain constraint to a friend? In my words, this means ___.	A block stays on a straight ramp. What is its acceleration perpendicular to the surface? I can tell because ___.
Example or sketch	**Non-example and why**
How would you use that fact to find the normal force when gravity and the surface are the only forces? My example shows ___.	Does zero perpendicular acceleration mean zero downhill acceleration? I would change ___ because ___.

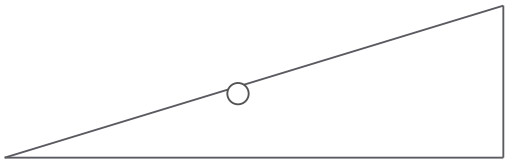
Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw the real weight once, then draw its two components in a separate box. Circle the component along the ramp.

Labels you can use: parallel · perpendicular · downhill.

Real forces



Components

Tell a partner

Along the ramp I use ___. Across the ramp I use ___.

One pull, two components

16

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Represent one force without counting it twice.

CHOOSE ONE JOB TOGETHER

NAVIGATION ENGINEER ☐

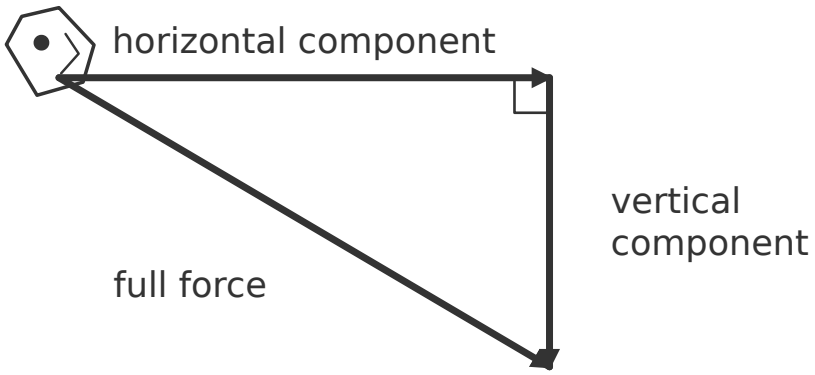
Prepare the component view for the calculator. Mark the full-force arrow that these components replace.

MAP GRAPHICS EDITOR ☐

Show the full force and its component view. Write a caption that prevents counting the same pull twice.

Word help: force / component / sum

A navigation graphic lists a gravitational force AND its two components as three separate pulls on XJ.



01 OUR FIRST CALL

Which arrows should enter the force sum? Explain your choice.

Use ___ because the components represent ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Check the calculator request. What correction would you send before it runs?

Now we would say ___ because ___.

Switch speakers. If XJ's velocity points horizontally, which force component changes its speed at that instant? Why?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks